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“Jnana Sangama”, Belagavi, Karnataka, India



A
Project Report
on

**“AUTOMATED SPEED MONITORING AND VEHICLE
CONTROL SYSTEM USING GPS”**

Submitted in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering
in
Electronics and Communication Engineering
Submitted By

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CERTIFICATE

Certified that the Project Entitled “**AUTOMATED SPEED MONITORING AND VEHICLE CONTROL SYSTEM USING GPS**” carried out by, **SPANDANA S PATIL**, bearing **USN 2JH22EC054** bonafide student of Jain College of Engineering and Technology, is in partial fulfillment for the award of the **BACHELOR OF ENGINEERING** in Electronics and Communication Engineering from **Visvesvaraya Technological University, Belagavi** during the year 2025-2026. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the report submitted in the department library. The Project report has been approved as it satisfies the academic requirements in respect of the Project prescribed for the said Degree.

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ABSTRACT

Contemporary society faces increasing challenges with traffic congestion and associated issues, particularly concerning vehicle speeding in restricted areas like school zones and hospitals. Manual policing and control in these areas is impractical, prompting the need for an automated solution. This project aims to create an Internet of Things smart GPS-based vehicle speed limit controller system in response to the rise in traffic accidents and growing concerns about road safety. The main goal is to increase road safety by using automated control systems and real-time data to enforce speed restrictions and reduce the number of accidents caused by speeding. This project takes a holistic approach, combining software and hardware elements. Sensors mounted on vehicles, such as GPS receivers and speed sensors, continuously collect data on the location and speed of the vehicle. Additionally, a variety of sources, including weather stations and traffic cameras, provide information on road conditions. A central control unit receives this data wirelessly, processes and analyses it. To compare vehicle speed limitations to predetermined speed limits for road segments, the control unit uses machine learning techniques. Depending on this analysis, it transmits control signals back to the vehicle when necessary. These signals can encompass driver warnings, power limitations, or autonomous speed control. By this in real-world scenarios, the system effectively mitigates speeding-related risks, leading to a substantial decline in traffic accidents. Its ability to adapt to changing road conditions and its scalability renders it a promising solution for future smart transportation systems.

Keywords: GPS-based Speed Control, Internet of Things (IoT), Machine Learning, Sensor Integration, Wireless Communication, Real-time Data Processing, Driver Assistance System, Speed Limit Enforcement.

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Chapter 1

INTRODUCTION

In recent years, traffic-related challenges have intensified at an unprecedented rate, primarily due to rapid urbanization, escalating vehicle density, and inadequately developed transport infrastructure. The uneven distribution of development and continuous population growth have further magnified these concerns, resulting in severe congestion and increased vulnerability to road accidents. Among the various causes of road mishaps, overspeeding remains one of the most dominant contributors, reflecting not only irresponsible driving behaviour but also a persistent neglect of fundamental road safety regulations. Such trends underscore the urgent need for innovative, technology-driven solutions capable of addressing the limitations of conventional traffic management mechanisms.

Traditional approaches to traffic monitoring often rely heavily on manual surveillance or expensive camera-based systems, many of which struggle to accurately detect dynamic variables such as pedestrian movement, lane-specific speed, or diverse environmental conditions. Despite technological advancements, ensuring real-time, high-precision road safety enforcement remains a significant challenge. In this context, the emergence of the Internet of Things (IoT), advanced sensor networks, and global positioning systems (GPS) has paved the way for highly efficient, automated, and intelligent transportation solutions.

One such promising innovation is the Smart GPS-Based Vehicle Speed Limit Controller System. This system represents a major paradigm shift from conventional methods by integrating real-time location tracking, automated speed regulation, and intelligent decision-making processes. It relies on empirical data and automated control mechanisms to minimize incidents caused by overspeeding, thereby offering a scientific, data-driven approach to road safety. Rather than remaining a conceptual model, this technology holds substantial practical relevance and demonstrates the potential to reduce accident rates, preserve human lives, and revolutionize the broader domain of smart transportation systems.

This research aims to explore the significance, development methodology, and projected impact of this system on enhancing road safety standards. Through empirical observations, technological analysis, and trend evaluation, the study seeks to highlight how intelligent speed-limiting solutions can contribute to safer roads and more efficient traffic management in an increasingly motorized world.

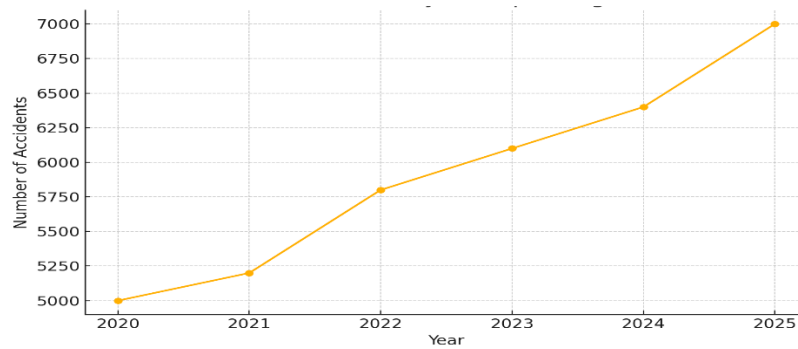


Fig 1.1.1- Road Accidents from Overspeeding (2020-2025)

The graph illustrates the trend in road accidents caused specifically by overspeeding during the period from 2020 to 2025. The X-axis represents the years within this range, while the Y-axis denotes the corresponding number of overspeeding-related accidents recorded annually. The plotted data indicates a consistent upward progression, beginning with 5000 incidents in 2020 and rising to 5200 in 2021. This is followed by a more substantial increase to 5800 in 2022. The numbers continue to escalate steadily, reaching 6100 in 2023 and 6400 in 2024, and ultimately peaking at 7000 accidents in 2025.

A clear and uninterrupted upward trend emerges from the graph, signifying a persistent growth in overspeeding-related road accidents over the six-year span. While the increase from 2020 to 2021 is relatively modest, the rise between 2021 and 2022 is notably significant, reflecting heightened risk factors during that period. From 2022 onward, the progression maintains a steady incline, indicating that overspeeding has become an increasingly prevalent and concerning issue. The year 2025 records the highest number of cases, underscoring the urgency of addressing speed-related violations more effectively.

This escalating pattern may be influenced by several interconnected factors. These include the increasing availability of high-performance vehicles capable of achieving greater speeds, inadequate enforcement of posted speed limits, and expanding urban environments that have not been matched with proportionate improvements in road infrastructure. Additionally, insufficient public awareness regarding the risks associated with overspeeding and limited deployment of automated surveillance or speed-monitoring technologies further contribute to the rising trend. Collectively, the data underscores the necessity for advanced, technology-driven safety solutions that incorporate real-time monitoring and proactive interventions to mitigate the growing threat posed by overspeeding on modern roadways.

1.1 PROBLEM STATEMENT

Increasing vehicle density, urban growth, and frequent overspeeding—especially near schools, hospitals, and residential areas—pose serious road safety challenges. Traditional methods like signboards, speed breakers, and manual policing depend on driver attention and are often ineffective. Even advanced systems such as cameras and radar only detect violations after they occur, offering no real-time prevention.

A major gap remains: the lack of an automated system that actively regulates vehicle speed based on GPS-defined restricted zones. Existing GPS alerts merely warn drivers but cannot enforce limits, allowing overspeeding due to negligence or rapid EV acceleration.

To address this, the project “Automated Speed Monitoring and Vehicle Control System Using GPS” proposes a real-time, cost-effective solution that identifies speed-limited zones through GPS and automatically controls vehicle speed. This proactive system enhances safety, reduces accidents, and contributes to intelligent transportation development.

1.2 OBJECTIVES OF THE PROJECT

- Designing a relay actuation circuit for motor control.
- Implementing GPS-based zone detection for restricted areas.
- Developing indication mechanisms such as speaker and TFT display for real-time alerts.
- Building a relay/motor control circuit for automatic speed limiting in restricted zones.

1.3 SCOPE OF THE STUDY

This study focuses on designing and implementing an automated speed monitoring and vehicle control system using GPS technology for predefined speed-restricted zones. The system detects the vehicle's location, identifies entry into restricted areas, and automatically regulates speed without driver input, ensuring safe operation in school zones, hospitals, residential areas, and accident-prone regions.

The work covers GPS-based geofencing, microcontroller programming, and electronic speed control suitable for electric vehicles. It examines how real-time data can be used to trigger automatic speed limitation and reduce human error or intentional overspeeding. The study also evaluates the system's accuracy, reliability, and performance under normal road conditions.

The scope is limited to fixed, predefined zones and does not include dynamic traffic updates or integration with official traffic databases. The system targets electric vehicles and focuses specifically on speed reduction rather than complete vehicle automation or collision-avoidance features. Despite these limits, the study provides a foundation for future developments in intelligent and automated transportation systems.

Chapter 2

LITERATURE SURVEY

Paper I

Modem Upendra, Varun Reddy and Dr. Safa M, “Smart GPS Based Vehicle Speed Limit Controller on zone identification using Geo-Fencing Algorithm” presented at 2024 3rd International Conference for Innovation in Technology (INOCON) Karnataka, India. Mar 1-3, 2024.

The authors present an IoT-enabled smart vehicle speed-limit controller that integrates GPS and a speed sensor to acquire real-time vehicle position and speed. Using a geo-fencing algorithm, the system identifies predefined zones, determines the applicable speed limit, and continuously evaluates whether the vehicle complies with it. A central control unit processes multi-source data, compares actual speed against zone limits, and issues warnings or automated control actions such as throttle restriction when overspeeding occurs. The system was tested both in controlled conditions and on public roads, achieving more than 96% accuracy in enforcing speed limits and demonstrating a notable reduction in speeding-related events.

Paper-II

Dakshata.R. Dhote, Pragati.R. Lakde, Anjali.S. Pachare, Mukesh.A. Meharkure, Mahendra Dongare, “Design and Implementation of Automatic Speed Control of Vehicle by Using RFID Technology” presented at International Journal of Research in Engineering, Science and Management Volume 4, Issue 7, July 2021.

The paper introduces an automatic vehicle speed-control system based on RFID technology for enforcing speed limits in restricted zones. RFID tags are positioned at the entry and exit points of a zone, while vehicles are equipped with an RFID reader connected to an ATmega328 microcontroller. When the reader detects a zone tag, the controller identifies the permitted speed and compares it with the vehicle's current speed. If overspeeding occurs, the system autonomously reduces speed without driver intervention. Although effective for localized control, the authors note limitations such as short sensing range and dependency on close RFID–reader alignment. The approach emphasizes a low-cost embedded solution aimed at reducing manual enforcement and improving safety in accident-prone areas.

Paper-III

Karthikeyan, Arshad Parvez, Kishore Kumar, Lavanya, Mathuprakas, “RFID-Based Automatic Vehicle Speed Control” presented at International Journal of Science, JETIR, Volume 10, Issue 1, January 2023.

The paper presents an RFID- and IoT-based automatic vehicle speed-control system designed to regulate speed in sensitive zones such as schools, hospitals, and congested areas. RFID tags are placed at zone boundaries, while vehicles carry an RFID reader connected to a microcontroller. When the reader detects a zone tag, the controller identifies the permitted speed, compares it with the vehicle's current speed, and automatically restricts speed if necessary. An IoT/WiFi module allows administrators to remotely configure zone-wise speed limits and monitor vehicle entries. The system emphasizes low-cost hardware and aims to improve enforcement in high-risk, speed-restricted zones.

Paper-IV

Kranthi Sai Reddy, Sai Jahnavi, Prathima Sheela, “Intelligent Smart Zone Based Vehicle Speed Control System Using RF” presented at Sreenidhi Institute of Science and Technology IJRECE Volume 6, Issue 4, October 2018.

The authors propose an RF-based smart-zone vehicle speed-control system to address overspeeding in sensitive areas such as schools, hospitals, and residential zones. RF transmitters placed at zone boundaries broadcast coded signals representing the zone type and its permitted speed. Vehicles are equipped with an RF receiver and microcontroller that detects these signals, identifies the zone entered, measures current speed, and automatically restricts speed if it exceeds the allowed limit. The system provides driver feedback through an LCD display indicating zone information and speed status. Designed as a low-range (≈ 60 m) wireless solution, it aims to improve safety and enforcement in high-risk areas through automatic, infrastructure-assisted speed regulation.

Paper-V

J. Aarthi Suganthi Kani, Amal P V, Monisha D, Shuchika S, Sarath Babu A S, “Zone Based Vehicle Speed Control System” presented at International Journal of Advances in Engineering and Management, IJAEM, Volume 5, Issue 2, February 2023.

The authors proposed a paper to control the speed of the vehicle automatically using the RF signal. Here the vehicle speed is controlled automatically without the help of a driver in the defined zones. Transmitters are installed at the boundaries of predefined zones, zones where speed regulation is required such as schools, hospitals, residential areas. These transmitters use RF technology radio frequency modules to broadcast a coded signal indicating the zone's identity and its associated speed-limit setting. Each vehicle is equipped with an RF receiver module that listens for the transmitter's signal when passing into the zone. The receiver connects to a microcontroller embedded system that monitors the vehicle's current speed via a speed sensor tied to a DC motor or vehicle drive in prototype and compares it to the predefined speed limit of the zone. When the vehicle enters the zone i.e., the transmitter signal is received, the system automatically reduces the vehicle's speed to the “cut-off” or permitted speed and maintains it throughout the zone until the vehicle leaves that defined area. After leaving the zone, control is returned to the driver and vehicle operates under normal traffic rules. The project uses a sensor to determine the vehicle's current speed, in the prototype this is a DC motor speed sensor. The microcontroller receives the sensor output, compares with the permitted limit. If actual speed > permitted, triggers the actuator/logic to reduce speed e.g., by controlling motor drive. The RF zone detection is effectively a trigger: when the vehicle enters the transmitter's range, the control module becomes active for speed limiting. The paper emphasizes the low-cost, integrated unit combining zone monitoring + vehicle control to enhance safety in high-risk areas.

HARDWARE DESCRIPTION

3.1 ESP-32 NODE MCU

The Node MCU ESP32 development board is a powerhouse of power and connectivity. With robust Wi-Fi and Bluetooth capabilities, it opens up a world of possibilities. Its compact 30-pin design allows it to seamlessly integrate into various products, enabling endless creative options. What sets this board apart are its dual wireless modes. You can choose between AP, STA, or AP+STA modes, giving you the flexibility you need for your IoT endeavors. At its core, the ESP32 Node MCU is powered by a dual-core Tensilica LX6 processor running up to 240 MHz. This gives it enough processing capability to handle real-time tasks, sensor data, and communication protocols simultaneously. It includes 520 KB of SRAM and supports external flash memory, ensuring smooth execution of larger programs. One of its most attractive features is its integrated Wi-Fi and Bluetooth both Classic and BLE connectivity, enabling wireless control and data exchange without additional hardware. In terms of connectivity and communication, the ESP32 Node MCU supports TCP/IP, HTTP, MQTT, and Web Socket protocols, making it an excellent choice for cloud-connected and real-time IoT systems. Its Bluetooth capabilities further expand its potential, allowing it to act as a Bluetooth beacon, connect to smartphones, or exchange data between nearby devices.



Fig 3.1.1 ESP32 NODE MCU

Features of ESP32 NODE:

- Powered by ESP32S chip
- Wi-Fi and Bluetooth capabilities
- 30-pin design for seamless integration
- Flexible AP, STA, or AP+STA modes
- CP2102 USB-UART Bridge

ESP32 NODE PIN DIAGRAM:

The ESP32 Node MCU board has 38 pins, which include power, GPIO (general-purpose input/output), ADC/DAC, touch, communication, and control pins. This flexibility allows the ESP32 Node MCU to connect with a wide variety of sensors, actuators, and modules such as temperature sensors, relays, displays, and motors.

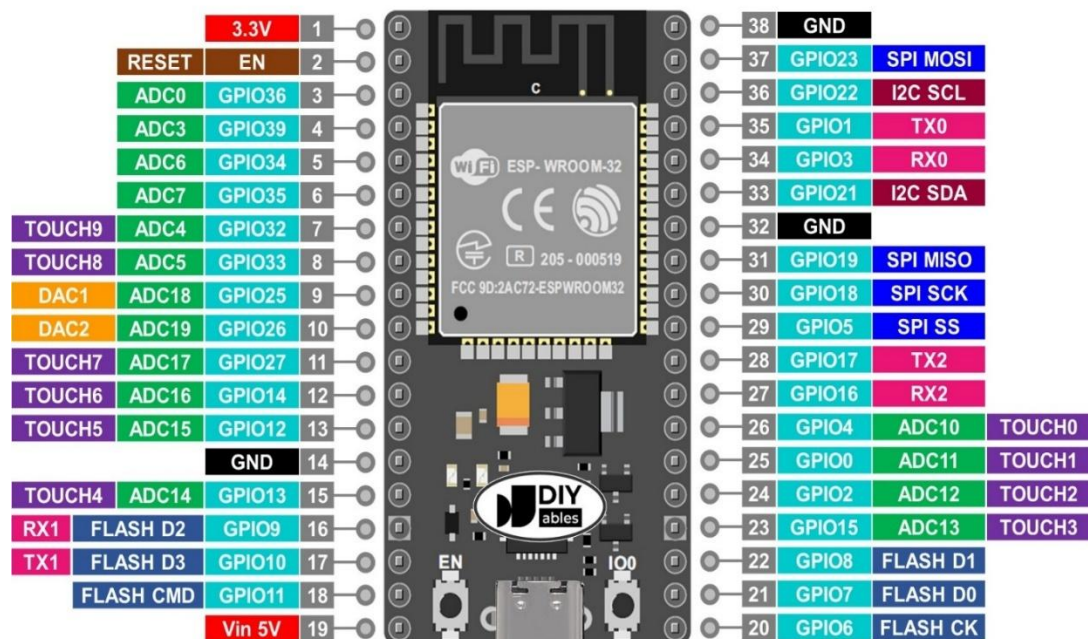


FIG 3.1.2 ESP32 Pin Diagram

Power Pins:

3.3v Pin1-Provides a stable 3.3V output to power external sensors or components. Maximum current is around 500mA. **Vin Pin19**-Input voltage (usually 5V from USB or external power supply). It powers the onboard voltage regulator that outputs 3.3V.

GND Pin14, 30, 38- Common ground for the circuit. Must be connected to all external device grounds.

Pin No.	GPIO	ADC	I/O	Special Functions
1	3V3	–	–	3.3V Power Output
2	EN	–	Input	Enable / Reset Pin
3	GPIO36	ADC1_CH0	Input only	Analog input
4	GPIO39	ADC1_CH3	Input only	Analog input
5	GPIO34	ADC1_CH6	Input only	Analog input
6	GPIO35	ADC1_CH7	Input only	Analog input
7	GPIO32	ADC1_CH4	I/O	RTC pin
8	GPIO33	ADC1_CH5	I/O	RTC pin
9	GPIO25	ADC2_CH8	I/O	DAC1, PWM
10	GPIO26	ADC2_CH9	I/O	DAC2, PWM
11	GPIO27	ADC2_CH7	I/O	PWM
12	GPIO14	ADC2_CH6	I/O	HSPI-CLK, PWM
13	GPIO12	ADC2_CH5	I/O	Boot, HSPI-MISO
14	GND	–	–	Ground
15	GPIO13	ADC2_CH4	I/O	HSPI-MOSI, PWM
16	GPIO9*	–	I/O	Flash — avoid
17	GPIO10*	–	I/O	Flash — avoid
18	GPIO11*	–	I/O	Flash — avoid
19	5V	–	–	USB 5V Input
20	GPIO6*	–	I/O	Flash SCK — avoid
21	GPIO7*	–	I/O	Flash SD0 — avoid
22	GPIO8*	–	I/O	Flash SD1 — avoid
23	GPIO15	ADC2_CH3	I/O	HSPI-CS, PWM
24	GPIO2	ADC2_CH2	I/O	Boot pin
25	GPIO0	ADC2_CH1	I/O	Boot mode
26	GPIO4	ADC2_CH0	I/O	RTC
27	GPIO16	–	I/O	UART2-RX
28	GPIO17	–	I/O	UART2-TX
29	GPIO5	–	I/O	SPI-CS, PWM
30	GPIO18	–	I/O	SPI-CLK, PWM
31	GPIO19	–	I/O	SPI-MISO, PWM
32	GND	–	–	Ground
33	GPIO21	–	I/O	I2C-SDA
34	GPIO3	–	I/O	UART0-RX
35	GPIO1	–	I/O	UART0-TX
36	GPIO22	–	I/O	I2C-SCL
37	GPIO23	–	I/O	SPI-MOSI, PWM
38	GND	–	–	Ground

Table 3.1.3 ESP32 Pin Description

Control Pins:

EN Pin2-Enable pin. When pulled LOW, it resets the ESP32. When HIGH, the board runs normally.

RESET -Used to restart the microcontroller manually. Often connected to the EN pin via a reset button.

Analog-to-Digital Converter (ADC) Pins-ESP32 has two ADC units (ADC1 and ADC2) with a 12-bit resolution (values from 0–4095).

3.2 NEO-6M GPS MODULE

A GPS module works by receiving signals from satellites, calculating distances to determine location using trilateration, and then outputting the position data, which can be used for navigation, tracking, and other applications. The NEO-6MV2 is a GPS (Global Positioning System) module and is used for navigation. The module simply checks its location on earth and provides output data which is longitude and latitude of its position. The NEO-6M module operates on the u-blox 6 GPS chipset, designed to offer fast satellite acquisition and tracking. It supports up to 22 tracking channels and 50 channels for acquisition, allowing it to receive data from multiple GPS satellites simultaneously. This helps the module determine its precise location (latitude, longitude, altitude) with an accuracy of about 2.5 meters under open-sky conditions.

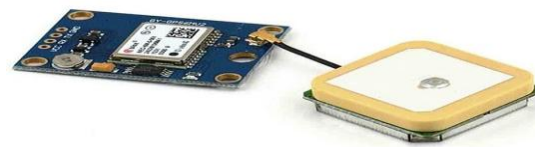


Fig 3.2.1 NEO 6M GPS Module

The NEO-6M GPS module is a highly efficient and easy-to-use device for obtaining location data. It provides accurate GPS readings and integrates smoothly with microcontrollers through UART communication. With its compact design and reliable performance, it has become one of the most widely used GPS modules in both academic and industrial applications.

Features:

- Location provided by it is accurate
- EPROM with battery back
- Supply Voltage: 3.3V Current Consumption: ~45mA
- Communication: UART (TTL logic)
- Position Accuracy: 2.5 meters
- Baud Rate: Default 9600 bps (configurable)

Applications:

- GPS-based tracking systems (vehicles, drones, and pets)
- Navigation devices.
- Geo fencing and location logging.

3.3 SIM800L GPRS GSM MODULE

The SIM800L GSM/GPRS module is a miniature GSM modem that can be used in a variety of IoT projects. You can use this module to do almost anything a normal cell phone can do, such as sending SMS messages, making phone calls, connecting to the Internet via GPRS, and much more. The module requires an external antenna in order to connect to the network. So the module usually comes with a helical antenna that can be soldered to it. The operating voltage of the chip ranges from 3.4V to 4.4V. The module supports baud rates ranging from 1200 bps to 115200 bps and features automatic baud rate detection.



Fig 3.3.1 SIM800L Module

The SIM800L is a quad-band GSM/GPRS module, which means it can operate on four different frequency bands: 850 MHz, 900 MHz, 1800 MHz, and 1900 MHz. This makes it compatible with most mobile networks worldwide. It communicates with a microcontroller via a UART (serial) interface using AT commands, a standard command set used to control GSM modems.

One critical aspect of the SIM800L is its power supply. It operates at 3.7V to 4.2V, similar to a Li-ion battery voltage. It cannot be powered directly from a 5V source like an Arduino 5V pin because it draws up to 2A of current during transmission bursts. Supplying inadequate current can cause the module to restart or fail to connect to the network. Hence, a dedicated 4V power source or a Li-ion battery with sufficient current capacity and decoupling capacitors e.g., 1000 μ F electrolytic capacitor is recommended.

Features:

- Supports Quad-band: GSM850, EGSM900, DCS1800 and PCS1900
- Connect to any global GSM network with any 2G SIM
- Make and receive voice calls using an external 8 Ω speaker & electret microphone
- Send and receive SMS messages
- Send and receive GPRS data (TCP/IP, HTTP, etc.)

Applications:

- IoT-based monitoring systems.
- Remote data logging and control.
- Vehicle tracking and security systems.
- SMS alert systems.
- Home automation with SMS commands.
- GPS + GSM combined trackers with NEO-6M

3.4 2.4-INCH TFT DISPLAY MODULE

The 2.4 display has 240X320 color pixels. The TFT driver (IL19341) can display full 16-bit color using library code. This 2.4inch TFT LCD Color Screen Module SPI Interface has 240X320 resolution, it uses the SPI interface to communicate with the controller such as Arduino, it is the best upgrading of the Nokia5110. Since the display uses 4-wire SPI to communicate and has its own pixel-addressable frame buffer, it can be used with every kind of microcontroller. This TFT (Thin Film Transistor) display uses LCD technology with an LED backlight, allowing it to show vivid colors (up to 65K or 262K colors, depending on the controller). The SPI interface enables communication between the display and a microcontroller such as Arduino, ESP32, or Raspberry Pi using only a few digital pins, making it efficient and easy to connect.



Fig 3.4.1 2.4-INCH TFT Display Module

The typical pin configuration includes VCC, GND, CS (Chip Select), RESET, DC/RS (Data/Command), SDI/MOSI (data input), SCK (SPI clock), and LED (backlight control). Some versions also feature touchscreen functionality, using a resistive touch sensor and a separate touch controller IC like the XPT2046. The SPI interface ensures that communication between the display and the microcontroller is efficient, even when using small 8-bit systems. For example, on an Arduino, the display can be connected using pins 11 (MOSI), 13 (SCK), 10 (CS), 9 (RESET), and 8 (DC).

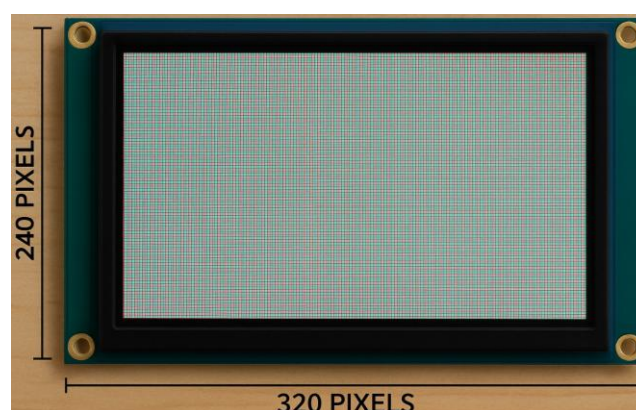


Fig 3.4.2 Display Pixel Dimensions

The 2.4-inch TFT module is widely used in embedded systems, Internet of Things (IoT) devices, and hobbyist electronics due to its affordability, color quality, and versatility. In embedded control systems, it serves as a human-machine interface (HMI) to display system status, menus, or data readings. In IoT applications, it can show sensor data, network status, or control options for smart devices. In DIY projects, it is ideal for digital clocks, game consoles, or graphical data loggers. Additionally, it finds use in industrial monitoring panels, portable test equipment, and educational projects that demonstrate SPI communication and graphics programming.

In summary, the 2.4-inch SPI 240×320 TFT display module is a compact, efficient, and visually rich display solution for modern embedded systems. Its combination of high resolution, vibrant color output, and compatibility with common microcontrollers makes it one of the most popular display modules for students, hobbyists, and engineers alike. With proper interfacing and software support, it enables the creation of sophisticated and interactive visual user interfaces in compact electronic designs.

Features:

- 2.4 -inch serial SPI color display module
- Support analog SPI and hardware SPI
- The LCD has a wide viewing angle, the contrast is also very suitable.
- Good quality display.
- Built-in white LED backlight for clear visibility.
- Brightness adjustable using PWM (Pulse Width Modulation).
- Handles pixel addressing, color data, and display refresh.
- Widely supported by Arduino, ESP32, STM32, and Raspberry Pi libraries.

Applications:

- Used in small embedded systems as a graphical user interface (GUI).
- Displays system parameters like temperature, speed, or voltage readings.
- Common in automation panels and control devices.
- Used in prototype smartwatches, fitness trackers, and small handheld devices.
- Compact and low power, making it suitable for battery-powered designs.
- Displays sensor data (temperature, humidity, pressure).
- Shows network information (Wi-Fi status, connectivity, etc.).

3.5 BTS7960 Motor Driver Module

The BTS7960 H-Bridge High-Power Motor Driver Module is a robust motor control board designed for driving large DC motors with high current demands. It uses two Infineon BTS7960 chips, which allow it to handle up to 43 amps of continuous current and voltages typically up to 24V or higher, depending on the application. This module provides efficient bidirectional motor control, allowing motors to move forward, reverse, stop, and brake smoothly. The built-in H-bridge configuration enables full control of motor direction and speed using PWM (Pulse Width Modulation) signals from a microcontroller such as an Arduino, Raspberry Pi, or ESP32.

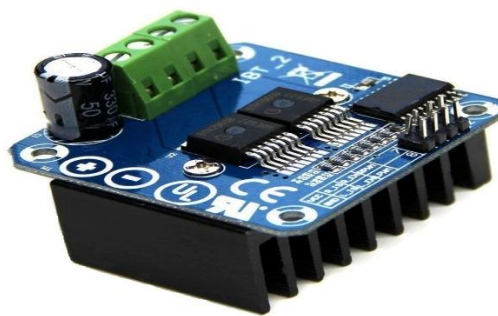


Fig 3.5.1 H-Bridge High Power Motor Driver

Features:

- Dual BTS7960 H-bridge driver chips for high current (up to 43A).
- Operating voltage: typically 6V–27V.
- Logic control voltage: 3.3V–5V, compatible with most microcontrollers.
- PWM speed control and direction control supported.
- Built-in overcurrent and thermal protection.
- High efficiency MOSFET-based design with low heat generation.
- Screw terminals for secure motor and power connections.
- LED indicators for status monitoring.

Applications:

- High-power DC motor control.
- Robotics (drive motors, robot arms).
- Electric vehicles, carts, and mobility projects.
- Conveyor systems and industrial automation.
- CNC machines or mechanical actuators requiring high torque.

3.6 Hall Effect Sensor Module

A Hall Effect Sensor Module is an electronic device used to detect magnetic fields. It works on the Hall Effect principle, which states that when a magnetic field is applied perpendicular to the flow of current in a conductor, a small voltage (called the Hall voltage) is generated across the conductor. This voltage can be measured and used to detect the presence, strength, or polarity of a magnetic field. Hall sensors are commonly used to sense motion, position, and speed, especially in rotating systems like motors, wheels, and fans.

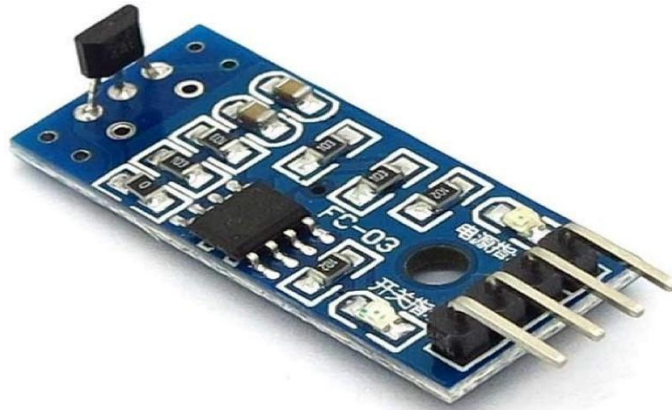


Fig 3.6.1 Hall Effect Sensor Module

Working Principle:

- When a magnet comes close to the sensor's detecting surface:
- The sensor produces an output signal (HIGH or LOW) depending on the magnetic field's presence or polarity.
- In digital Hall sensors, the output switches ON/OFF.
- In analog Hall sensors, the output voltage varies with magnetic field strength.

Features:

- Operating Voltage: 3.3V to 5V (compatible with Arduino, Raspberry Pi, etc.)
- Digital or Analog Output: Depending on the model.
- Built-in LED indicator for signal detection.
- High sensitivity and fast response time.

Applications:

- Speed and position sensing in DC motors and fans.
- Revolution counters or tachometers.
- Contactless switch (e.g., door sensor, proximity sensor).
- Magnet-based encoder systems.

3.7 MP3 MODULE

The MP3-TF-16P MP3 SD Card Module with Serial Port is a compact and inexpensive MP3 module that can be connected directly to the speaker. Module with battery power supply, speaker, the keypad can be used alone, or through the serial port control, as the Arduino UNO or any microcontroller with a serial port module. The module itself perfectly integrated hardware decode MP3, WAV, WMA. While the software supports TF card driver to support FAT16, FAT32 file system. The MP3-TF-16P module is a plug-and-play MP3 player with a micro-SD card interface, ideal for embedded projects requiring simple audio playback. Its multiple control pins and standard SD card compatibility make it highly versatile for hobbyists and engineers alike.

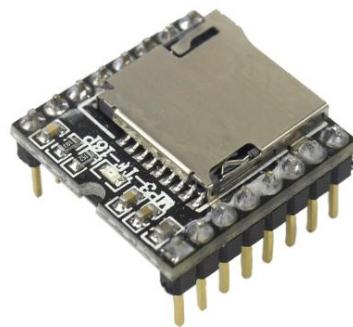


Fig 3.7.1 MP3 Module

How It Works:

- Insert a micro-SD card with properly formatted audio files (MP3/WAV).
- Power the module with a suitable voltage (3.3V or 5V).
- Control playback via the hardware buttons connected to control pins or via serial communication (if supported).
- The module decodes MP3 files and outputs analog audio signals on speaker outputs.
- Connect speakers or an audio amplifier to these outputs for sound playback.

Features:

- Supports micro-SD cards: Usually supports up to 32GB micro-SD cards formatted as FAT16 or FAT32.
- MP3 and WAV playback: Can play common audio file formats stored on the SD card.
- 16-pin interface: Provides pins for power, audio output, control, and communication.
- Audio output: Usually stereo output pins (left and right channels) for connection to speakers or amplifiers.

3.8 Speaker

A 0.5W 80 Ω trumpet speaker is a small, horn-shaped sound device used in alarms, toys, and electronic circuits. It has a power rating of 0.5 watts, meaning it can handle moderate sound levels suitable for low-power applications. The 80-ohm impedance limits current flow, making it ideal for sensitive circuits or signal devices. The trumpet shape helps amplify and direct sound efficiently. Its diameter determines tone—larger sizes produce deeper sounds, while smaller ones give sharper tones. Compact and efficient, this type of speaker is perfect for producing clear, focused audio output in compact or portable electronic systems.



Fig 3.8.1 Speaker

Features:

- Power Rating: 0.5 watts – suitable for low-power audio or alarm applications.
- Impedance: 80 ohms – ensures low current draw and good compatibility with small circuits.
- Trumpet Shape: Enhances sound projection and clarity.
- Compact Design: Small and lightweight, ideal for portable or embedded devices.
- Clear Sound Output: Produces focused, audible tones for alerts and notifications.
- Durable Construction: Built with sturdy materials for reliable performance.
- Easy Installation: Simple two-wire connection for quick setup.
- Wide Application: Used in alarms, buzzers, toys, and small electronic gadgets.

3.9 DC MOTOR

A DC motor (Direct Current motor) is an electromechanical device that converts electrical energy (DC) into mechanical rotational energy. It operates on the fundamental principle of electromagnetic induction-when a current-carrying conductor is placed in a magnetic field, it experiences a force that causes motion.

Inside a DC motor, there are two main parts:

- Stator (stationary part): Produces a magnetic field using either permanent magnets or electromagnets.
- Rotor or Armature (rotating part): A coil of wire that carries current and rotates due to the interaction with the stator's magnetic field.



Fig 3.9.1 DC Motor

When a DC voltage is applied, current flows through the armature winding, creating an electromagnetic force (EMF). According to Fleming's left-hand rule, this force produces torque, causing the shaft to rotate. The speed and direction of the motor can be controlled by adjusting the input voltage or using a motor driver (like an H-bridge circuit). DC motors are widely used because they offer precise speed control, high starting torque, and simple operation.

Features:

- Operates on direct current (DC) supply.
- High starting torque, suitable for heavy-load starting conditions.
- Smooth and precise speed control through voltage or PWM control.
- Simple design and easy maintenance.
- Reversible rotation by changing polarity of supply voltage.
- Available in multiple types: brushed, brushless (BLDC), geared DC motors, etc.
- Compact and efficient for small to high-power applications.
- Can operate on batteries, making it ideal for portable devices.

3.10 Potentiometer

A potentiometer (pot) with a knob is a three-terminal variable resistor used to adjust voltage or current in a circuit manually. By rotating the knob, you change the resistance between the wiper (middle terminal) and the two end terminals, allowing smooth control over electrical devices such as volume, brightness, or motor speed.

The knob makes it easy to adjust precisely, providing a tactile and visual way to control the output. Potentiometers can be linear (resistance changes proportionally to rotation) or logarithmic (changes follow a curve, often used in audio applications).



Fig 3.10.1 Potentiometer

Features:

- Three terminals: two fixed ends and one wiper.
- Adjustable resistance over a specified range.
- Smooth rotary control with a knob for manual adjustment.
- Compact and easy to mount on boards or panels.
- Available in linear or logarithmic taper.
- Durable design for repeated use.
- Can be used in analog circuits for fine-tuning voltage or signal levels.

Applications:

- Volume control in audio devices.
- Brightness control for LEDs or lamps.
- Speed control for DC motors.
- Sensor calibration in analog circuits.
- Adjustable voltage dividers in electronics experiments.
- User interfaces in DIY projects or control panels.

3.11 Lithium-Ion Battery

A Lithium-Ion 2S Single Cell Battery consists of two lithium-ion cells connected either in series (2S), the voltage doubles to about 7.4V while the capacity remains the same. In parallel (2P), the voltage stays around 3.7V, but the capacity doubles. These batteries offer high energy density, lightweight design, and rechargeable capability. They are widely used in portable electronics, drones, power banks, and DIY projects. Proper protection circuits (BMS) are essential to prevent overcharging, over-discharging, and short circuits, ensuring safety and extending battery life. They require compatible chargers and careful handling to avoid damage.



Fig 3.11.1 Lithium-Ion Battery

Features:

- **High energy density:** More energy storage in a compact size compared to NiMH or NiCd batteries.
- **Nominal voltage:** About 3.7V per cell.
- **Rechargeable:** Can be charged hundreds of times.
- **Low self-discharge:** Retains charge for a longer time when not in use.
- **Lightweight:** Ideal for portable electronics.
- **Protection required:** Needs protection circuitry for safe operation (overcharge, over-discharge, short-circuit protection).

3.12 Mini Step-down Converter

The Mini 360 is a compact DC-DC step-down (buck) converter that efficiently converts a higher input voltage to a lower, regulated output voltage. Unlike linear regulators, it is a switching regulator, which makes it highly efficient (often 90% or higher), generating less heat while delivering higher currents. It's widely used in DIY electronics, robotics, and battery-powered projects to provide stable voltages for microcontrollers, sensors, LEDs, and other devices.



Fig 3.12.1 Mini Step-down Converter

Working Principle:

- The Mini 360 uses a switching MOSFET and inductor to step down voltage.
- By rapidly switching the input voltage on and off and smoothing it through an inductor and capacitor, it produces a stable lower voltage.
- The output voltage is adjustable via a small onboard potentiometer.
- Input voltage typically ranges from 4.5V to 28V, and output can be set from about 1.25V up to 20V depending on the module.

Features:

- High efficiency (up to 90–95%), unlike linear regulators.
- Adjustable output voltage with a small onboard potentiometer.
- Compact and lightweight, ideal for breadboards and small projects.
- Input voltage range: 4.5V – 28V.
- Output current: up to ~2A (with proper cooling).
- Built-in short-circuit, overcurrent, and thermal protection.
- Low ripple output suitable for sensitive electronics.

Applications:

- Powering Arduino, ESP32, Raspberry Pi, and other microcontrollers.
- Battery-powered electronics requiring regulated voltages.
- Converting higher battery voltages (like 12V or 24V) to lower voltages for circuits.

3.13 Buzzer

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.



Fig 3.13.1 Buzzer

The pin configuration of the buzzer is shown above. It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-' symbol or short terminal and it is connected to the GND terminal.

Specifications:

- Color is black
- The frequency range is 3,300Hz
- Operating Temperature ranges from -20°C to $+60^{\circ}\text{C}$
- Operating voltage ranges from 3V to 24V DC
- The sound pressure level is 85dBA or 10cm
- The supply current is below 15mA

Applications:

The applications of the buzzer include the following.

- Communication Devices
- Electronics used in Automobiles
- Alarm Circuits
- Portable Devices
- Security Systems
- Timers
- Household Appliances

Chapter 4

SOFTWARE DESCRIPTION

4.1 ARDUINO IDE

The Arduino Integrated Development Environment (IDE) is a crucial software tool for programming and uploading code to Arduino microcontroller boards. It offers a user-friendly interface where developers can write, compile, and debug their code. The IDE supports the Arduino programming language, which is based on C/C++, and includes built-in functions for controlling hardware components, such as LEDs, sensors, and motors. This open-source platform is compatible with various operating systems, making it accessible and widely used in electronics prototyping, automation, and IoT projects.



Fig 4.1.1 Arduino IDE Software

Arduino Functions:

The functions allow a programmer to divide a specific code into various sections, and each section performs a particular task. The functions are created to perform a task multiple times in a program.

The function is a type of procedure that returns the area of code from which it is called. For example, to repeat a task multiple times in code, we can use the same set of statements every time the task is performed.

Advantages of using functions:

- It increases the readability of the code.
- It conceives and organizes the program.
- It reduces the chances of errors.

- It makes the program compact and small.
- It avoids the repetition of the set of statements or codes.
- It allows us to divide a complex code or program into a simpler one.
- The modification becomes easier with the help of functions in a program.

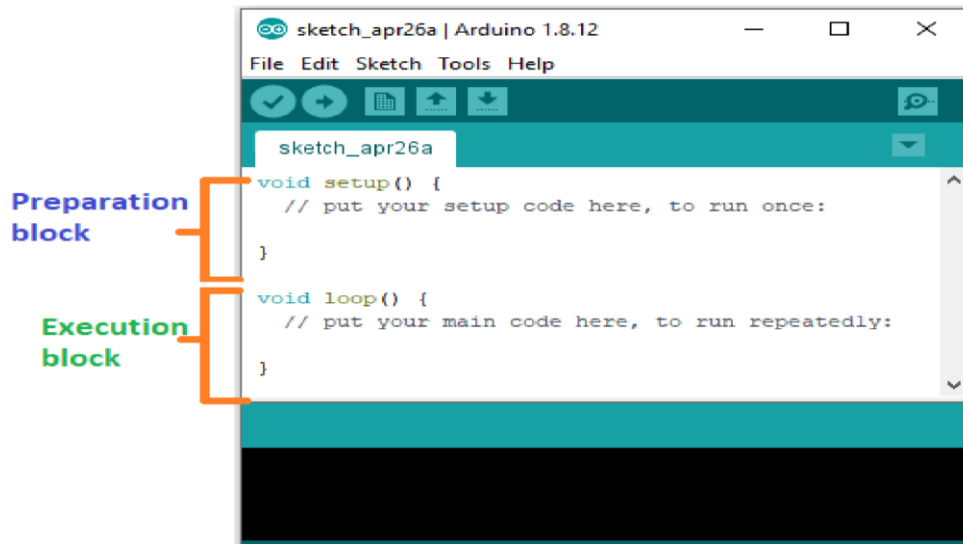


Fig 4.1.2 Arduino IDE Software Execution Example

4.2 JSONBin.io

JSONBin.io is a cloud-based platform designed to store, manage, and retrieve JSON (JavaScript Object Notation) data. It serves as a lightweight and flexible online database solution for developers who require an easy-to-use backend for applications, prototypes, and IoT systems. JSONBin.io provides RESTful API endpoints that allow data to be accessed programmatically from any device, including microcontrollers such as ESP32 or Arduino.

JSONBin.io has become popular due to its simplicity, quick setup, and ability to provide real-time data storage without the need to deploy dedicated servers or large database systems. It allows developers to create “bins” where JSON data is securely stored and can be updated or retrieved whenever needed.

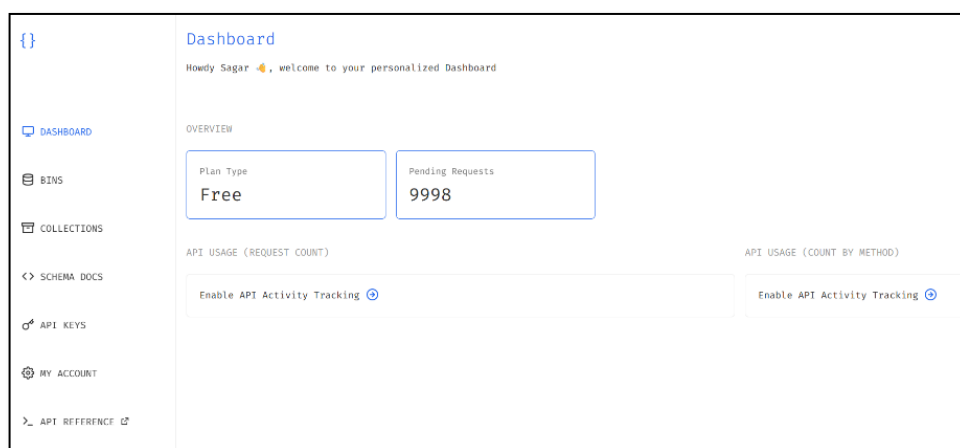


Fig 4.2.1 JSONBin.io Interface

JSONBin.io works as a cloud-based JSON host where each JSON file is stored as a Bin. A bin is essentially a data container that holds structured JSON. Users can access these bins through URLs generated by the platform.

It is commonly used in:

- Web Application Development
- IoT and Embedded Systems (ESP32, ESP8266, Arduino)
- Mobile App Development
- Automation Systems
- Student Academic Projects

JSONBin.io is especially beneficial for projects requiring remote monitoring, real-time updates, or cloud-based data logging.

Working Principle:

The working of JSONBin.io revolves around HTTP communication using REST APIs.

Developers can:

- POST data to create a new JSON bin
- PUT data to update existing JSON
- GET data to retrieve the current version of the bin
- DELETE a bin when no longer needed

The platform also supports versioning, meaning each update is stored as a new version, allowing previous versions to be accessed when required.

Features of JSONBin.io:

1. Cloud-Based JSON Storage

JSONBin.io removes the need for maintaining a private server or database. Users can directly store JSON files on the cloud and access them whenever needed.

2. RESTful API Support

Each JSON bin can be accessed through API endpoints, making integration easy with applications, microcontrollers, mobile apps, or websites.

3. Data Versioning

Every time data is updated, JSONBin.io automatically creates a new version. This helps maintain a history of changes and improves debugging.

4. Public and Private Modes

Users can make bins publicly accessible or restrict access using secure keys. This ensures safe data sharing and enhances privacy.

5. High Availability

The cloud infrastructure ensures that data is always accessible with minimal downtime. This reliability is crucial for IoT and automation projects.

6. Simple and User-Friendly Interface

The dashboard provides an easy way to create and manage bins. Even beginners can use JSONBin.io without deep backend knowledge.

Advantages of JSONBin.io:

- Simple to Use: No database management needed.
- Cost-Effective: Free tier available for student projects.
- API-Based: Works seamlessly with JavaScript, Python, ESP32, Android, etc.
- Secure Access: API keys protect sensitive data.
- Structured Data Format: JSON is easy to parse and widely supported.
- Version History: Helps track modifications over time.
- High Performance: Fast response times for both GET and POST requests.

Applications:

- IoT Device Monitoring
- Weather Stations
- GPS Tracking Projects
- Smart Home Automation
- Vehicle Monitoring Systems
- Mobile Application Data Hosting
- Online Configuration Storage
- Data Backup for Small Apps

METHODOLOGY

5.1 FLOW CHART

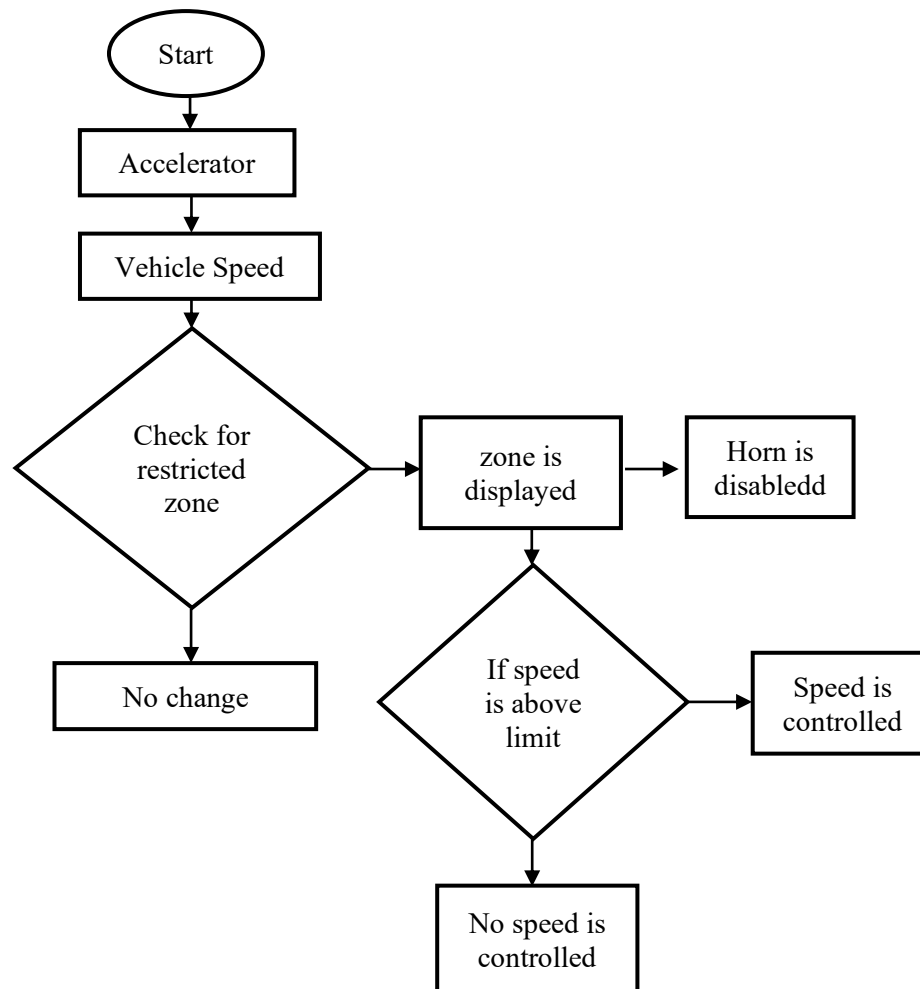


Fig 5.1.1 Flow Chart of Vehicle Speed Control System

This flowchart represents a methodology for monitoring and controlling vehicle speed, especially in restricted zones. Here is a step-by-step explanation of the methodology:

1. Start: The process begins.
2. Accelerator Input: The system takes input from the accelerator to determine vehicle speed.
3. Measure Vehicle Speed: The current speed of the vehicle is recorded.
4. Check for Restricted Zone: The system checks whether the vehicle is in a restricted zone.

- If yes, the process moves to the next step.
- If no, there is no change, and the vehicle continues moving normally.

5. Restricted Zone Actions:

- If in a restricted zone, the system displays the speed.
- The Horn is disabled in the restricted zone.

6. Speed Control:

- The system checks if the vehicle's speed is above the allowed limit.
- If yes, the speed is controlled (possibly by limiting acceleration).
- If no, there is no speed control, and the vehicle continues as usual.

7. Purpose:

This methodology ensures that vehicles comply with speed limits in restricted zones.

Disabling the horn prevents unnecessary noise pollution in designated areas.

Automatic speed control enhances safety by preventing over speeding.

5.2 BLOCK DIAGRAM

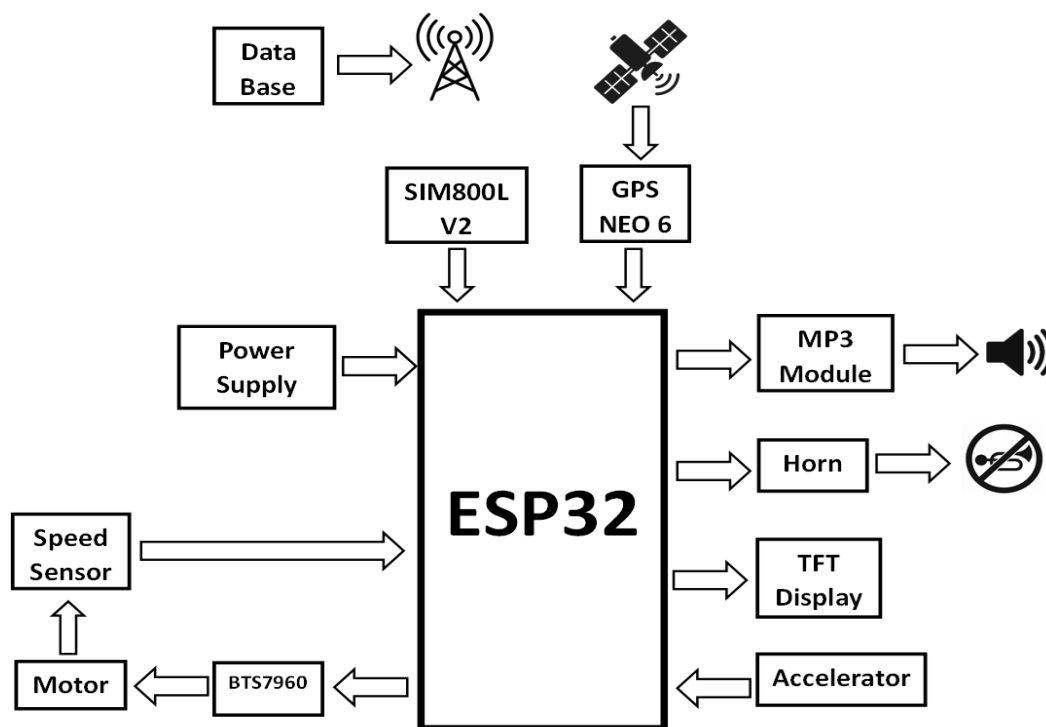


Fig 5.2.1 Block Diagram

The block diagram represents the integrated architecture of the Automated Speed Monitoring and Vehicle Control System. The ESP-32 microcontroller functions as the central processing unit, interfacing with multiple sensing, communication, and actuation modules. The GPS NEO-6 receiver provides continuous geospatial data, enabling real-time tracking of latitude, longitude, and satellite availability essential for geofencing and zone detection. The SIM800 GSM module enhances system connectivity by offering cellular communication capability for remote alerts and data transmission. A Hall-effect-based speed sensor monitors wheel rotation and generates pulses proportional to the vehicle's velocity, which the ESP-32 converts into calibrated speed information.

The accelerator input is conditioned and processed to determine the user-defined throttle level, subsequently fed into the motor driver, which modulates power delivered to the DC motor. Output peripherals such as the TFT display, horn, and MP3 audio module are controlled by the ESP-32 to deliver real-time feedback, enforce no-honking zones, and broadcast audio warnings during zone entry and overspeed events. A regulated power supply ensures stable operation of all subsystems. Collectively, the block diagram delineates a tightly coupled cyber-physical system in which sensing, computation, and actuation operate in synchrony to achieve intelligent vehicle safety management.

5.3 CIRCUIT DIAGRAM

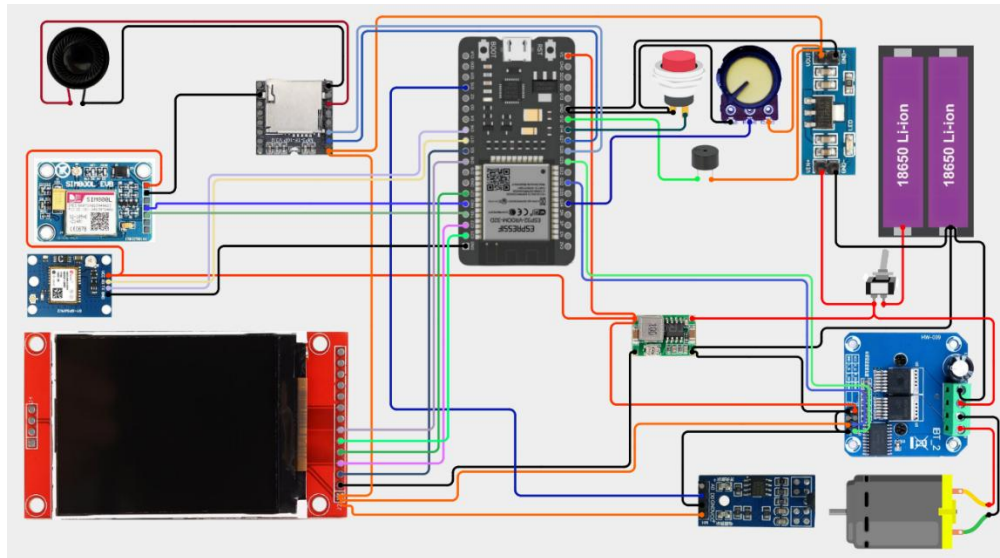


Fig 5.3.1 Circuit

Component	Pin Name	ESP32 Pin
GPS	TX	16 (RX1)
GPS	RX	17 (TX1)
GPS	VCC	3.3V/5V
GPS	GND	GND
TFT	CS	5
TFT	DC	21
TFT	RST	22
TFT	MOSI	23
TFT	SCK	18
TFT	MISO	Not used
TFT	VCC	5V
TFT	GND	GND
SIM800L	TXD	GPIO3 (RX0)
SIM800L	RXD	GPIO1 (TX0)
SIM800L	VCC	External 5V 2A
SIM800L	GND	Common GND
Hall Sensor	OUT	15
Hall Sensor	VCC	3.3V
Hall Sensor	GND	GND
BTS7960	RPWM	25
BTS7960	R_EN	32
BTS7960	LPWM	Not used
BTS7960	L_EN	Not used
Potentiometer	Output	34
Potentiometer	VCC	3.3V
Potentiometer	GND	GND
Horn Button	Signal	12
Horn Buzzer	Control	14

Table 5.3.2 Complete Pin Mapping

5.4 RESTRICTED ZONE DATABASE

The Restricted Zone Database forms the core geospatial dataset used by the Automated Speed Monitoring and Vehicle Control System for intelligent zone classification and speed regulation. This database stores predefined geofenced regions, each represented by its corresponding latitude–longitude coordinates, permitted speed limit, and operational constraints such as horn restrictions. The ESP32 continuously acquires real-time positioning data from the NEO-6M GPS module and computes the vehicle’s current geo-location. Using an in-built geofencing algorithm, the system checks whether the live coordinates fall within the defined boundaries of any restricted zone.

Upon detection of zone entry, the microcontroller triggers a zone-specific control routine. For school, hospital, and residential zones, the algorithm enforces strict speed limiting by modulating the PWM signal sent to the BTS7960 motor driver, thereby capping the maximum achievable speed. Additionally, horn-disabling logic is activated in silence-critical areas to minimize noise pollution. The TFT display dynamically updates zone information, speed thresholds, and alert messages, while the DFPlayer module delivers audible notifications to ensure driver awareness.

Outside restricted zones—primarily highways and open roads—the control logic reverts to normal operation, allowing higher speed limits and re-enabling horn control. This structured database-driven approach provides deterministic, automated, and real-time enforcement of road-safety rules, reducing dependence on human judgment and minimizing the possibility of overspeeding or non-compliance.

The table below summarizes the geofenced zones incorporated into the system prototype:

Zone Name	Latitude	Longitude	Speed Limit	NO-Honking
Hospital Zone (KIMS)	15.361031	75.126900	30 kmph	True
KLE Banyan School	15.372542	75.117900	20 kmph	True
Unkal Cross	15.372907	75.117492	40 kmph	False
Campus Zone (KLE)	15.370938	75.118546	30 kmph	True
Campus Zone (JCET)	15.394204	75.119950	30 kmph	True
Sai Nagar Circle	15.382313	75.117674	40 kmph	False
Rani Channamma Circle	15.350236	75.137617	40 kmph	False
Market	15.345726	75.142880	40 kmph	False

Table 5.4.1 Zone Information Data Base

To visually represent the spatial distribution of these zones, a geofenced map is generated, as shown in Fig. 5.4.2. Each marker indicates the centroid of a predefined control zone; these reference points are used by the ESP32's geofencing subroutine to trigger appropriate speed-limiting and alert mechanisms.

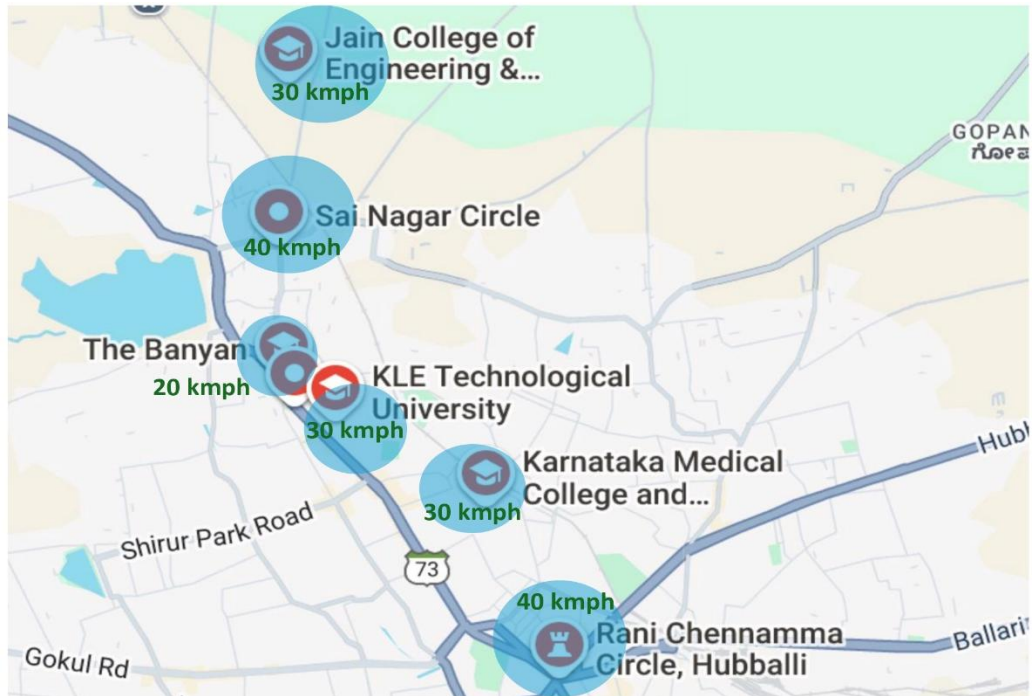


Table 5.4.2 Location Map of Defined Speed Control Zones

RESULTS AND DISCUSSION

6.1 FINAL OUTPUT MODEL



Fig 6.1.1 Final Output Model



Fig 6.1.2 Restricted Zone & NO-Zone Condition

6.2 RESULTS

The final output of the project is a Smart Vehicle Control System that automatically monitors and regulates vehicle speed using GPS-based zone detection and an electronic throttle-control mechanism. The system continuously reads GPS data, identifies whether the vehicle has entered a predefined speed-restricted zone, and automatically limits the motor speed based on zone-specific speed limits.

A TFT display shows real-time information such as latitude, longitude, throttle level, horn status, current speed, zone name, and allowed speed limit. When the vehicle enters a restricted area, the system plays an audio alert through the DFPlayer module and prevents overspeeding by capping the throttle and gradually reducing speed.

The model also includes a no-horn zone feature, automatically disabling the horn in silent areas. Hall sensor feedback ensures accurate speed measurement, while the relay/PWM motor control provides reliable and smooth speed regulation. The system operates without driver input, ensuring consistent, automatic, and real-time speed control for enhanced road safety in critical zones.

6.3 ADVANTAGES

1. Road accidents due to over-speeding especially in crowded regions can be majorly avoided.
2. Significantly increases road safety.
3. Speed is controlled automatically no manual intervention is needed.

6.4 LIMITATIONS

- GPS accuracy drops in tunnels, indoors, or dense city areas.
- SIM800L uses 2G, which is slow and may have poor network coverage.
- System depends on internet to download updated geofence data.
- No encryption in 2G GPRS communication, so data security is limited.

6.5 APPLICATIONS

1. **Safety:** Helps to reduce the number of accidents and injuries in areas like schools, hospitals, and sharp turns.
2. **Ease of driving:** Makes driving easier and safer in traffic.
3. **Automatic control:** Takes control of the vehicle's speed automatically when it enters a restricted zone
4. **Low cost:** Can be low-cost, long-lasting, and low-power
5. **Reliable communication:** Can provide reliable communication to authorized society in case of an emergency.
6. **It can be implemented two wheelers (EV's).**
7. **This system can be implemented in IC vehicles.**
8. **It can also be implemented in highways.**

6.6 APPROXIMATE COST

Sl No	Component	Rate	Qty	Total
1	ESP32-38 Pin Development Bord	1000	1	1000
2	SIM800L V2 Wireless GSM GPRS Module	750	1	750
3	NEO-6M GPS Module with EPROM	700	1	700
4	2.4-inch SPI Interface 240x320 TFT Display	850	1	850
5	BTS7960B 43A H-Bridge Motor Driver	800	1	800
6	12V DC Motor	150	1	150
7	Hall Effect Sensor + Neodymium Magnet	150	1	150
8	MP3 Module & Speaker	280	1	280
9	2S Battery Holder & Li-ion Battery	260	1	260
10	Mini Step-Down Buck Converter Module	150	2	300
11	5V Passive Buzzer	30	1	30
12	10K Potentiometer with Knob	30	1	30
14	6x4 PCB	100	1	100
15	Labor (body frame cutting & drilling)	500	1	500
16	Travel (Visit to various road location to test the model)	500	1	500
17	Miscellaneous Components (Wire, Solder, Glue, Switches, Connectors, Acrylic Sheet, Wheel, Gear, Shaft)	2100	1	2100
		Total		8,500/-

Table 6.6.1-Project Cost Approximation

CONCLUSION

This project successfully implements a GPS-enabled, zone-aware vehicle speed regulation system using the ESP32 platform. By integrating geofencing, Hall-sensor-based speed measurement, and PWM-driven throttle control, the system autonomously enforces speed limits within predefined regions. The model demonstrates reliable real-time detection of restricted zones, automatic speed limiting through a model-mapped control algorithm, and effective communication of system status via TFT display and audio alerts. The inclusion of no-horn zone enforcement further enhances safety functionality. Overall, the system provides a technically robust foundation for intelligent transportation applications and can be extended with dynamic traffic data, cloud integration, or advanced vehicle automation in future work.

FUTURE SCOPE

The Smart Vehicle Monitoring and Control System provides a strong base for advancing intelligent transportation technologies. Its modular architecture allows several future enhancements:

- 1. Integration with Smart Traffic Infrastructure** - Future versions can interface with centralized traffic management servers, enabling real-time uploading of vehicle speed, GPS data, and zone violations. This would support automated monitoring, hotspot analysis, and dynamic traffic regulation.
- 2. Mobile Application Support** - A dedicated Android/iOS app can provide live vehicle tracking, geofence alerts, overspeed logs, and remote configuration of speed limits and zones, enhancing usability for drivers and fleet operators.
- 3. Hybrid Positioning for Offline Accuracy** - Incorporating IMU-based dead-reckoning (accelerometer, gyroscope, magnetometer) can maintain accurate location and speed estimation in GPS-challenged environments such as tunnels or dense urban areas.
- 4. 4G/5G IoT Connectivity** - Upgrading to modern modules like SIM7600 or EC25 can improve reliability and scalability. LPWAN technologies such as NB-IoT or LTE-M can further support large-scale deployments.

REFERENCES

- [1] Modem Upendra, Varun Reddy, and Dr. Safa M, “Smart GPS Based Vehicle Speed Limit Controller on Zone Identification Using Geo-Fencing Algorithm,” presented at the *2024 3rd International Conference for Innovation in Technology (INOCON)*, Karnataka, India, Mar. 1–3, 2024.
- [2] Dakshata R. Dhote, Pragati R. Lakde, Anjali S. Pachare, Mukesh A. Meharkure, and Mahendra Dongare, “Design and Implementation of Automatic Speed Control of Vehicle by Using RFID Technology,” *International Journal of Research in Engineering, Science and Management*, vol. 4, no. 7, pp. 126–130, Jul. 2021.
- [3] Karthi Eyan, Arshad Parvez, Kishore Kumar, Lavanya, and Mathupraas, “RFID-Based Automatic Vehicle Speed Control,” *JETIR – Journal of Emerging Technologies and Innovative Research*, vol. 10, no. 1, pp. 450–455, Jan. 2023.
- [4] Kranthi Sai Reddy, Sai Jahnavi, and Prathima Sheela, “Intelligent Smart Zone Based Vehicle Speed Control System Using RF,” *IJREE – International Journal of Research in Electronics and Communication Engineering*, vol. 6, no. 4, pp. 215–220, Oct. 2018.
- [5] J. Aarthi Suganthi Kani, Amal P. V., Monisha D., Shuchita S., and Sarath Babu A. S., “Zone Based Vehicle Speed Control System,” *International Journal of Advances in Engineering and Management (IJAEM)*, vol. 5, no. 2, pp. 110–114, Feb. 2023.
- [6] S. Prakash, M. Hariharan, and R. Manoj Kumar, “GPS and IoT Based Vehicle Speed Regulation and Zone Monitoring System,” in *Proc. International Conference on Smart Electronics and Communication (ICOSEC)*, Tamil Nadu, India, Sep. 2022.
- [7] A. Bhatt, P. Desai, and R. Shah, “Geo-Fencing Enabled Speed Control System for Intelligent Transportation,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 11, pp. 250–254, Nov. 2020.
- [8] Harshita Sharma and Deepak Gupta, “Microcontroller Based Automatic Speed Limiter Using GPS,” presented at the *International Conference on Computing, Communication and Automation (ICCCA)*, Noida, India, 2021.

- [9] S. Khan, M. Shaik, and V. Chary, “Smart Vehicle Speed Limiting System Using IoT and Cloud Integration,” *International Journal of Electrical and Electronics Engineering (IJEEE)*, vol. 13, no. 3, pp. 45–52, Jun. 2022.
- [10] R. Patel and A. Joshi, “Real-Time Speed Control and Monitoring System for Electric Vehicles Using GPS and Embedded Systems,” presented at the *IEEE International Conference on Sustainable Energy and Intelligent Systems (SEIS)*, Hyderabad, India, 2023.
- [11] P. Sharma and R. Verma, “IoT-Based Intelligent Vehicle Speed Monitoring and Alert System Using GPS and GSM,” *International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)*, vol. 10, no. 8, pp. 9876–9882, Aug. 2021.
- [12] N. Singh and S. Tiwari, “Smart Road Safety and Speed Regulation System Using Embedded Controllers,” in *Proc. 2022 International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, Mar. 2022.
- [13] M. R. Alvarado and L. J. Garcia, “Geo-Fence Based Vehicle Speed Restriction System for Urban Transport,” *IEEE Access*, vol. 9, pp. 156234–156241, 2021.
- [14] P. K. Sahu and A. Mishra, “Design of GPS–GSM Based Vehicle Overspeed Detection and Auto Notification System,” in *Proc. 2023 International Conference on Intelligent Systems and Green Technology (ICISGT)*, Kerala, India, Apr. 2023.
- [15] A. R. Mohammed and S. Daniel, “Automated Vehicle Speed Control Using IoT and Real-Time Tracking,” *International Journal of Computer Applications (IJCA)*, vol. 183, no. 25, pp. 28–34, Nov. 2021.